SURJECTIVITY ON THE F-SQUARED KERNELS SPLITS THE ETA TRANSFORM INTO A DIRECT SUM OF SUBMODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author writes 'We omit the verification that this is well defined; the assumption of the lemma exactly signifies that the domain of this operation is all of Z^i.' inside the counted proof span; that sentence is kept verbatim. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The eta transform of a complex of f-torsion free modules and the associated short exact sequence. invoked at proof lines 27822-27823 and 27830-27831. Supplies the objects being split and the sequence whose splitting is constructed.
- The identification of the hypothesis with surjectivity of the f-squared kernel onto the f-kernel. used at proof lines 27826-27828. Makes the map t^{-1} defined on all of Z^i.
- Compatibility of the resulting direct sum decomposition with the displayed canonical map. used at proof lines 27832-27836. Is the conclusion that the canonical map identifies the source with a direct sum of submodules.

Source reference. Original remark 27579: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27579–27579. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a ring and let $f \in A$ be a nonzerodivisor. Let $M^\bullet$ be a complex of f-torsion free A-modules. If* $\mathrm{Ker}(d^i \bmod f^2)$ *surjects onto* $\mathrm{Ker}(d^i \bmod f)$*, then the canonical map*

$$(1, d^i) : (\eta_f M)^i / f(\eta_f M)^i \longrightarrow f^i M^i / f^{i+1} M^i \oplus f^{i+1} M^{i+1} / f^{i+2} M^{i+1}$$

identifies the left hand side with a direct sum of submodules of the right hand side.

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Proof. With notation as in Remark 27579 we define a map $t^{-1} : Z^i \to (\eta_f M)^i/f(\eta_f M)^i$. Namely, for $x \in M^i$ with $d^i(x) = f^2 y$ we send the class of x in Z^i to the class of $f^i x$ in $(\eta_f M)^i/f(\eta_f M)^i$. We omit the verification that this is well defined; the assumption of the lemma exactly signifies that the domain of this operation is all of Z^i. Then $t \circ t^{-1} = \mathrm{id}_{Z^i}$. Hence t^{-1} defines a splitting of the short exact sequence in Remark 27579 and the resulting direct sum decomposition

$$(\eta_f M)^i/f(\eta_f M)^i = Z^i \oplus B^{i+1}$$

is compatible with the map displayed in the lemma. □